

**A NOVEL ANTENNA DESIGN FOR SATELITE
RE-ENTRY VEHICLE FOR MITIGATION
COMMUNICATION BLOCKOUT
A CAPSTONE PROJECT REPORT**

Submitted by

**Y.BINDU PRIYA (9922005077)
D.MASTHAN VALI (9922005097)
B.AKHILESWARA REDDY (9922005345)**

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Dr. P.MANIKANDAN.,M.E., Ph.D.



**DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING
SCHOOL OF ELECTRONICS, ELECTRICAL AND BIOMEDICAL TECHNOLOGY
KALASALINGAM ACADEMY OF RESEARCH AND EDUCATION
KRISHNANKOIL**

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SCHOOL OF ELECTRONICS, ELECTRICAL AND BIOMEDICAL TECHNOLOGY
DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

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This is to declare that this project is our original work. No part of this work has been submitted elsewhere partially or fully for the award of any other degree or diploma. Relevant previous works presented in this report have been properly acknowledged and cited. The plagiarism policy, as stated by the project guide and project coordinator, has been maintained.

REGISTER NUMBER	STUDENT NAME	STUDENT SIGNATURE
9922005077	Y.BINDU PRIYA	_____
9922005097	D.MASTHAN VALI	_____
9922005345	B.AKHILESWARA REDDY	_____

Date: _____

KALASALINGAM ACADEMY OF RESEARCH AND EDUCATION
SCHOOL OF ELECTRONICS, ELECTRICAL AND BIOMEDICAL TECHNOLOGY
DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

BONAFIDE CERTIFICATE

Certified that this capstone project report on “**A NOVEL ANTENNA DESIGN FOR SATELITE RE-ENTRY VEHICLE FOR MITIGATION COMMUNICATION BLOCKOUT**” is the work of Y.BINDU PRIYA (9922005077) , D.MASTHAN VALI (9922005097) and B.AKHILESWARA REDDY (9922005345) who carried the work under my supervision (Dr. P.MANIKANDAN.,M.E., Ph.D.).

Dr. P.MANIKANDAN.,M.E., Ph.D.
(PROJECT GUIDE)

Dr. CHARLES PRAVIN J.,M.E., Ph.D.
(HEAD OF THE DEPARTMENT)

Submitted for the Capstone Project Viva-voce held on _____

Internal Examiner

External Examiner

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ABSTRACT

During atmospheric re-entry, spacecraft experience an intense aerodynamic heating process that produces a high-temperature ionized gas layer, commonly known as a plasma sheath. This plasma significantly affects electromagnetic wave propagation and has historically led to communication blackouts in several missions, limiting real-time telemetry, tracking, and guidance. Numerous studies have attempted to mitigate this issue through techniques such as high-power transmitters, frequency shifts, aerodynamic shaping, and protective materials; however, these methods often suffer from high complexity, increased system weight, or inconsistent performance under varying plasma densities. Recent research trends increasingly explore advanced electromagnetic structures and material-based solutions to improve signal penetration through plasma, yet challenges remain in achieving reliable, low-loss communication during the most critical phases of re-entry. This work builds upon these existing gaps by examining the behaviour of antenna systems under plasma-affected conditions and evaluating potential enhancements that can improve overall electromagnetic performance. The study focuses on understanding how plasma alters antenna characteristics and explores practical approaches that may contribute to reducing communication degradation. The findings aim to support future developments in re-entry communication technologies and provide guidance toward minimizing blackout periods while ensuring mission safety.

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LIST OF ABBREVIATIONS

ACRONYM	DEFINITION
EM	Electro Magnetic
RF	Radio Frequency
CST	Computer Simulation Technology
SMA	SubMiniature Version A (Connector)
S_{11}	Reflection Coefficient / Return Loss
VSWR	Voltage Standing Wave Ratio
GHz	Gigahertz
MM	Metamaterial
MNG	Mu-Negative Metamaterial
ENG	Epsilon-Negative Metamaterial
SRR	Split Ring Resonator
PEC	Perfect Electric Conductor
PMC	Perfect Magnetic Conductor
B.H.	Blackout (Re-entry Plasma Communication Blackout)
EVA	Effective Vehicle Antenna
RCS	Radar Cross Section
FZP	Fresnel Zone Plate
LORENTZ	Lorentz Dispersion Model
PLASMA	Ionized Gas Layer around Re-entry Vehicle
FDTD	Finite Difference Time Domain
ROI	Region of Interest
dB	Decibel
MHz	Megahertz

CHAPTER 1 INTRODUCTION

1.1 Background and Motivation

During spacecraft reentry, a high-temperature plasma sheath forms around the vehicle, causing radio blackout by reflecting communication signals. Traditional solutions like magnetic windows or low-frequency antennas are bulky or inefficient. A metamaterial-based antenna design offers a lightweight, tunable alternative. By pairing a mu-negative metamaterial layer with the plasma's epsilon-negative behaviour, it enables signal transmission through the plasma. This approach enhances real-time communication during reentry, addressing critical telemetry and safety needs [2]. This metamaterial design can blend smoothly into the spacecraft's heat shield, staying strong even in extreme temperatures. Since it works passively, it doesn't need extra power, making it more efficient. Its ability to adjust to changing plasma conditions helps maintain steady communication, improving safety, control, and reliability during the most critical phase of reentry.

1.2 Problem Statement

During spacecraft reentry, the formation of a dense plasma sheath around the vehicle causes a communication blackout by blocking or reflecting radio frequency (RF) signals. Existing mitigation methods such as magnetic windows or low-frequency transmissions are limited by size, power, and bandwidth constraints. There is a need for a compact, efficient, and adaptable solution that allows continuous, real-time communication through the plasma environment. This challenge becomes increasingly critical for modern spacecraft and hypersonic vehicles that rely on uninterrupted telemetry, navigation, and control links. The blackout not only disrupts mission data flow but also compromises crew safety and autonomous operations. Emerging metamaterial-based antenna technologies offer a promising alternative by manipulating electromagnetic properties to counter plasma effects. Developing such systems could enable seamless signal transmission, reduce system complexity, and mark a significant advancement in reentry communication reliability and aerospace mission safety [3].

1.3 Objectives and Outcomes

- To design a metamaterial-based antenna system capable of mitigating communication blackout through the plasma layer.
- To evaluate the tunability and adaptability of the metamaterial design under varying plasma densities and reentry conditions.
- To simulate and validate the effectiveness of metamaterial structures (e.g., mu-negative Meta surfaces) in creating pass-

bands through epsilon-negative plasma environments. • To propose a practical and reliable antenna solution for enhancing space mission safety and communication stability. • Analyse Plasma Effects on Communication: Understand how high-temperature plasma sheaths formed during reentry affect radiofrequency (RF) signal propagation, including reflection, absorption, and signal loss.

1.4 Scope and Applications

This project focuses on designing and analysing a metamaterial-based patch antenna to overcome communication blackouts caused by plasma during spacecraft reentry. It explores the integration of epsilon-negative (ENG) and mu-negative (MNG) metamaterials to manipulate electromagnetic waves for improved signal transmission. The study emphasizes antenna design, simulation under realistic plasma conditions, and performance evaluation. While experimental validation is beyond this scope, the research provides valuable insights into passive, compact, and efficient mitigation strategies. This work lays the foundation for future practical implementation in spacecraft communication systems.

1.5 Applications

Satellites and Probes: Enhances data transmission during critical phases of atmospheric entry. Space Missions: Ensures continuous communication between spacecraft and ground stations during re-entry, improving mission safety and reliability. Defense and Aerospace: Can be adapted for high-speed aircraft and defense systems facing plasma effects. Satellite and Deep-Space Communication: Can be adapted to reduce signal loss in plasma-rich environments like ionospheres or planetary atmospheres. Hypersonic Vehicles: Supports real-time control and monitoring of high-speed atmospheric vehicles where plasma-induced signal loss is common.

CHAPTER 2 STATE OF ART AND NEED

2.1 Existing Research, Technologies or Products

Early investigations documented long, mission-critical communication blackouts during re-entry and explored higher-frequency telemetry and fluid injection as partial mitigations. NASA reviews summarize aerodynamic, magnetic, and active-injection approaches and underline trade-offs in mass, power, and feasibility. More recent theoretical and numerical work shows that layered metamaterial/meta surface structures—alternating epsilon-negative (ENG) and mu-negative (MNG) layers or optimized meta surfaces—can enable electromagnetic coupling through an ENG-like plasma sheath and thereby reduce attenuation at certain bands; simulation studies report reduced return loss and improved transmitted power in low-to-moderate plasma densities. Experimental and lab studies exploring plasma photonic crystals and controlled plasma columns demonstrate proof-of-concept wave transmission mechanisms, but full flight-level validation remains limited [5].

2.1.1 *Research Literature*

During spacecraft re-entry, extreme heat ionizes atmospheric gases, forming a plasma layer around the vehicle. This plasma, rich in free electrons, reflects and absorbs radio frequency (RF) signals, causing a communication blackout between the spacecraft and ground stations. The blackout typically lasts a few seconds to minutes, disrupting telemetry and control. This poses serious risks to mission safety and data transmission. Understanding plasma blackout is crucial for developing technologies that maintain reliable communication during re-entry. Metamaterial-based research explores engineered materials with negative permittivity (ENG) or permeability (MNG), enabling unique control over electromagnetic waves. These materials are applied in antenna design to counter plasma-induced signal loss during spacecraft re-entry. Studies show that metamaterial-based antennas improve signal transmission, bandwidth, and communication stability, effectively mitigating plasma blackout and enhancing re-entry communication reliability [6].

2.1.2 *Existing Technologies or Commercial Products*

Existing plasma blackout mitigation technologies include magnetic windows, which use strong magnetic fields to reduce plasma density and improve signal transmission; extended probe antennas, physically distancing antennas from dense plasma layers; chemical quenchers that inject gases to lower plasma density; and frequency selection, using higher-frequency signals that better penetrate plasma. However, these methods often face challenges like

added weight, complexity, limited effectiveness, or integration difficulties during hypersonic reentry [6].

2.1.3 *Relevant Patents*

- Plasma Antenna — US 6,492,951 B1 (Assignee: The Regents / Inventor: T. Anderson). Describes electrodeless plasma tubes used as reconfigurable radiators where a plasma column forms the antenna element. Relevant as a direct plasma-based radiator concept; differs from our passive metamaterial approach which does not require onboard plasma generation.
- Plasma Antenna System and Method — US 2003/0142021 A1. Covers drive and control methods for plasma antennas, including optical/laser-driven plasma. Highlights implementations and control electronics that increase system complexity—an important contrast to passive metamaterial layers.
- RF Transmission Through a Plasma — US 3,758,862 A. Early patent describing impedance matching and adaptive tuning to maintain RF transmission through ionized gas. Useful background on impedance-matching strategies but different from ENG/MNG metamaterial coupling methods.
- Metamaterial Antenna — WO 2013/060115 A1 (and related family). Claims antenna designs that integrate metamaterial layers to improve performance and miniaturization. Directly relevant to implementation techniques for ENG/MNG layers and mechanical mounting on curved surfaces. Careful claim review is needed to avoid overlap in unit-cell geometry or claimed layer stacking methods.
- Antenna Structure with Metamaterial — US 11,005,187 B2. Describes flexible metamaterial layers attachable to vehicle surfaces and passive magnetic-property element arrays—highly relevant for conformal metamaterial layers on re-entry vehicles. Note restrictions around specific element shapes and mounting approaches.
- Plasma Switched Array Antenna — US 10,230,166 B2. Covers arrays using plasma switches for reconfiguration. Illustrates active reconfigurable topologies (plasma as switch) that contrast with passive ENG/MNG meta surfaces; useful when considering reconfigurable vs. passive trade-offs.

2.2 **Gaps and Challenges in Existing Approaches**

Limited flight validation: Many mitigation concepts (metamaterials, magnetic windows, fluid injection) are supported mainly by simulations or ground-lab tests; full flight-representative validation at realistic re-entry plasma densities and temperatures is scarce. This reduces confidence in operational readiness.

- Mass, power and complexity trade-offs: Active approaches such as magnetic biasing or plasma generation require heavy coils, high power, or plumbing for injectants, which conflict with strict spacecraft mass/power budgets and increase failure modes.
- Thermal and material robustness: Re-entry exposes surfaces to extreme temperatures, ablation and erosion; proposed metamaterial layers and metasurfaces often lack demonstrated thermal resilience, conformal attachment methods, or survivability

under thermal-mechanical stress. • Bandwidth and angle sensitivity: Plasma properties vary with frequency, incidence angle, and vehicle geometry. Some solutions work only in narrow bands or specific incidence conditions, limiting reliable telemetry across required channels. • Fabrication and integration challenges: Designing lightweight, conformal ENG/MNG unit cells at high temperatures and integrating them with thermal protection systems (TPS) without degrading antenna performance remains understudied. • IP and practical deployment risks: Patent filings cover active plasma antennas and some metamaterial topologies, so novel designs must avoid claim overlap and address manufacturability and certification hurdles. . These gaps motivate a low-mass, passive, thermally robust metamaterial-integrated antenna approach that is validated with realistic plasma models and focused on manufacturable, conformal solutions.

2.3 Need for the Proposed Project

spacecraft re-entry communication is severely affected by plasma-induced blackouts, posing risks to telemetry, navigation, and crew safety. Existing solutions like magnetic windows, fluid injection, and active plasma antennas are often heavy, complex, and mostly validated only in simulations or laboratory settings . Metamaterial-based antennas show promise in simulations for improving signal transmission and reducing losses under plasma conditions, but practical validation under realistic plasma densities, thermal stresses, and operational constraints is limited. This project addresses these gaps by exploring a lightweight, passive, and thermally robust metamaterial antenna to enhance reliable communication during re-entry.

CHAPTER 3 REQUIREMENTS AND CONSTRAINTS

3.1 Requirements

3.1.1 Antenna Requirements

- The antenna must operate at **2.4 GHz**.
- The designed patch antenna in free space must exhibit a return loss (S11) of approximately **-20 dB**.

3.1.2 Antenna Under Plasma Environment

- The plasma layer should **block communication at 2.4 GHz** due to its epsilon-negative behaviour
- Expected return loss should degrade to approximately **-10 dB** as plasma prevents signal penetration

3.1.3 Antenna under plasma with MNG Metamaterial

- The combined system must operate at **2.4 GHz**.
- The metamaterial (ENG/MNG) layer should **enable communication even in the presence of plasma**.
- Return loss should improve back to approximately **-20 dB** after metamaterial compensation

Definition: Functional requirements describe *what the system must do*, including supporting communication, blocking signals with plasma, and restoring transmission with ENG metamaterial.

Nature:

- Binary: either satisfied or not.
- Directly linked to electromagnetic behaviour and project objectives.
- Essential for proving plasma-blockout mitigation.

3.1.4 Performance Requirements

Definition: Performance requirements describe *how well the system performs*, including S11 values, gain, bandwidth stability, and radiation characteristics under plasma.

- **Return Loss (S11):**
 - Free space: approx. **-20 dB**.
 - Plasma: approx. **-10 dB** (communication blocked).
 - Antenna + Plasma + ENG: restored to approx. **-20 dB**.
- **Radiation Pattern Stability:** The antenna should maintain a stable radiation pattern even under plasma influence.
- **Metamaterial Effectiveness:** ENG/MNG SRR array must provide a passband through plasma's epsilon-negative region.
- **Frequency Stability:** Operating frequency should remain locked at **2.4 GHz** in all environments.

Table 3.1: Summary of Requirements

Req. ID	Description	Category	Target / Expectation
FR1	Patch antenna must resonate at 2.4 GHz	Functional	Operating frequency = 2.4 GHz
FR2	Plasma should block signal propagation	Functional	S11 degraded to approx. -10 dB
FR3	ENG metamaterial must restore communication	Functional	S11 returns to approx. -20 dB
PR1	Maintain return loss performance	Performance	Free space = -20 dB; Plasma = -10 dB; ENG = -20 dB
PR2	Radiation pattern stability	Performance	Minimal distortion under plasma
PR3	Plasma penetration capability	Performance	Demonstrated EM coupling through plasma

3.2 Engineering Standards and Codes

- **IEEE Std 145-2013:** Standard definitions of antenna terms—used for S11, gain, VSWR and radiation parameter definitions.
- **IEEE Std 149-2021:** Standard test procedures for antennas—referenced for validating antenna performance under plasma and free-space conditions.
- **IEEE Std 1128-1998:** Methods for antenna near-field measurements—relevant for analysing field interactions with metamaterial structures.
- **IEEE Std 299-2006:** Standard for electromagnetic shielding effectiveness—supports comparison of plasma attenuation effects.

- **IEEE Std 1720-2012:** Recommended practice for near-Earth space systems—relates to communication challenges during re-entry.
- **IEEE Std 1589-2014:** Standard for space system communications and configuration—used for understanding re-entry communication requirements.
- **IEEE Std 1528-2013:** EM field exposure and power absorption—relevant to safety considerations during testing.
- **IEEE Std 1900.6-2011:** Spectrum sensing interfaces and data structures—applicable to dynamic plasma environments affecting RF propagation.

Table 3.2: Summary of Standards Considered

Standard	Domain / Scope	Relevance to Project
IEEE 145-2013	Antenna terminology and definitions	Provides standard definitions for S11, gain, radiation efficiency, and other antenna parameters used throughout the project.
IEEE 149-2021	Antenna test procedures	Ensures accurate evaluation of antenna performance, including S-parameters and radiation patterns under plasma and free-space conditions.
IEEE 1128-1998	Near-field antenna measurement methods	Relevant for analyzing field distribution around metamaterial structures and plasma-affected surfaces.
IEEE 299-2006	Electromagnetic shielding effectiveness	Supports comparison of plasma's EM shielding effect and validates metamaterial-based mitigation.
IEEE 1720-2012	Near-Earth space systems	Provides guidelines related to communication challenges during atmospheric re-entry.
IEEE 1589-2014	Space system communications	Supports understanding of re-entry communication requirements and blackout mitigation strategies.

3.3 Design Constraints

3.3.1 Thermal Constraints

- During atmospheric re-entry, temperatures can exceed **1800–2000°C**, creating a highly ionized plasma sheath.
- The antenna must be conceptually compatible with materials that can withstand these extreme thermal loads.
- Metamaterial layers (MNG) designed so that their electromagnetic performance remains predictable under high-temperature plasma environments.

3.3.2 *Frequency and Bandwidth Constraints*

- The design focuses on the **2–5 GHz** band, typically used for spacecraft telemetry and communication links.
- Plasma frequency and collision frequency must not overlap with the intended operating band to avoid complete signal attenuation.
- Bandwidth must remain stable even when plasma density varies during different re-entry phases.

3.3.3 *Simulation Constraints*

- Accurate modelling requires defining plasma parameters such as electron density, collision frequency, and plasma thickness.
- CST Studio Suite supports plasma modelling, but simplified Drude-based plasma models limit real-world accuracy.
- Computational constraints restrict simulation of full vehicle geometry, requiring planar or simplified antenna–plasma models.

3.3.4 *Size Constraints*

- Patch antenna and metamaterial layers must fit within the limited surface area available on a re-entry vehicle.
- Metamaterial split-ring resonators (SRR) must maintain sub-wavelength dimensions while still offering MNG behaviour.
- The overall structure should be thin and lightweight to avoid interfering with thermal protection system (TPS) tiles.

3.4 **Data Ethics and Management**

Not applicable: This project does not involve datasets, personal information, or user data. Only simulation models were used, so no data privacy, storage, or ethical concerns arise.

CHAPTER 4 DESIGN AND IMPLEMENTATION

4.1 Concept Development and Alternatives

This section presents the different antenna design approaches examined to mitigate plasma-induced communication blackout during spacecraft re-entry. Each concept was evaluated based on performance, feasibility, robustness, and compatibility with re-entry plasma conditions.

4.1.1 *Concept A – Conventional Microstrip Patch Antenna*

This concept uses a standard rectangular microstrip patch antenna placed on a dielectric substrate (FR-4 or Rogers). It serves as the baseline reference.

Advantages:

- Simple structure and easy to fabricate.
- Low cost and minimal design complexity.

Drawbacks:

- Severe signal attenuation under plasma sheath conditions.
- No mechanism to counter plasma reflection or absorption.
- Narrow bandwidth and unstable radiation in high-density plasma.

4.1.2 *Concept B – Slot-Loaded Multiband Patch Antenna*

This concept introduces slots (U-slot, E-slot, or circular slot) to extend bandwidth and achieve multiband behavior.

Advantages:

- Supports multiband operation (useful for aerospace standards).
- Improved impedance matching and bandwidth.
- Better gain compared to the simple patch.

Drawbacks:

- Still experiences strong S11 degradation during plasma blackout.
- Radiation efficiency drops significantly when electron density increases.
- Cannot recover signal below the blackout threshold (−10 dB).

4.1.3 Concept C – Metamaterial-Based Antenna (Proposed Design)

This concept integrates a mu-negative (MNG) metamaterial layer or an SRR-based metasurface beneath the patch to manipulate electromagnetic wave propagation through the plasma sheath.

Advantages:

- Reduces plasma-induced reflection and absorption.
- Provides negative permittivity/permeability, enabling wave tunneling.
- Significant improvement in S11, gain, VSWR, and radiation stability.
- Effective in restoring communication below blackout threshold.

Drawbacks:

- Requires precise CST optimization to tune metamaterial parameters.
- More complex fabrication compared to standard patches.
- Limited real-world testing for extreme re-entry plasma conditions.

4.1.4 Evaluation Criteria

The following criteria were used to evaluate and compare the concepts:

Table 4.1: Evaluation Criteria for Antenna Concepts

Criteria	Description
Cost	Material and fabrication cost for practical implementation.
Feasibility	Ease of modelling, CST simulation, and prototype realization.
Performance	S11 improvement, gain enhancement, VSWR stability, and plasma transmission ability.
Robustness	Stability under varying plasma dielectric parameters (electron density, collision frequency).
Compliance with Standards	Compatibility with aerospace frequency bands and IEEE antenna design standards.

4.2 Final Design and Justification

Based on the detailed trade-off analysis carried out in Chapter 4, three alternative concepts were evaluated for mitigating plasma-induced communication blackout during spacecraft re-entry. The results clearly indicate that **Concept C – Patch Antenna Integrated with**

Metamaterial (SRR-based MNG layer) provides superior electrical performance and satisfies plasma penetration requirements.

4.2.1 Summary of Design Alternatives

- **Concept A – Conventional Patch Antenna:** Provided four resonant bands in free space but failed completely under plasma loading. Plasma frequency exceeded antenna operating frequency, causing complete signal attenuation.
- **Concept B – Patch Antenna with Bandwidth Enhancement:** Modified geometry improved bandwidth but did not improve plasma penetration. Plasma still caused S11 to approach 0 dB, resulting in communication loss.
- **Concept C – Patch Antenna with SRR-Based MNG Metasurface:** Introduced negative-permeability (MNG) layer using split-ring resonators. Enabled ENG–MNG tunneling effect that restored resonances under plasma. Achieved 4 usable bands with S11 < –10 dB even in plasma environment.

The designed antenna achieved 4 resonant bands: **2.42 GHz, 3.32 GHz, 4.24 GHz, 4.78 GHz**. Best match observed at 2.42 GHz with **S11 = –36 dB** in free space.

Conclusion: Although Table 4.2 and Table 4.3 both evaluate the design options, only one table is required for assessment. Based on the results, **Concept C was selected as the final design** because it provides the only technically valid solution capable of transmitting RF signals through a plasma sheath.

4.2.2 Novelty of the Work

This project introduces several novel contributions compared to existing plasma blackout mitigation techniques:

- Unlike traditional antennas that fail completely under plasma loading, the proposed design integrates a **mu-negative (MNG) metasurface**, enabling wave tunneling through the plasma's epsilon-negative (ENG) behavior.
- Existing literature primarily studies individual metamaterial layers; this project develops a **custom SRR lattice tuned to 2.4–5 GHz re-entry communication bands**.
- A complete simulation-based framework was developed using **CST Microwave Studio with Drude plasma model**, enabling realistic evaluation of plasma-electromagnetic interactions.

Table 4.2: Comparison of Alternative Solutions

Criteria	Concept A (Patch Antenna)	Concept B (Bandwidth Enhanced Patch)	Concept C (Metamaterial-Based Antenna)
Performance in Plasma	Poor – all resonances lost	Moderate – only improved	Excellent – ENG/MNG tunneling restores communication
Complexity	Low	Medium	High (metasurface design + integration)
Plasma Penetration Capability	None	Very Low	High – S11 restored below –10 dB
Suitability for Re-entry Vehicle	Not suitable	Partially suitable	Best choice – meets mission requirements
Overall Assessment	Weak	Moderate	Highest technical performance

Table 4.3: Trade-off Analysis of Design Alternatives

Criteria	Weight	Concept A	Concept C
Plasma Penetration	0.35	1 (no penetration)	5 (ENG–MNG tunneling)
S11 Performance	0.25	3	5
Feasibility	0.20	4	3
Integration with Heat Shield	0.10	3	4
Manufacturability	0.10	4	3
Weighted Score	1.00	2.45	4.35

- The novelty lies in integrating:

Patch Antenna + SRR Metasurface + Plasma Model

as one coupled system, which is rarely explored in undergraduate-level or even early research-level works.

- The proposed metasurface produces **double-negative (DNG) behavior** when interacting with plasma, significantly reducing mismatch losses.

This establishes the originality and research significance of the antenna design.

4.2.3 IP Considerations

Given the unique geometry and integration method, the system has strong intellectual property potential:

- The **SRR arrangement and tunability method** can be filed as a hardware patent under metamaterial antenna innovations.
- The **layer stacking approach** for plasma mitigation does not directly overlap with existing active plasma antenna patents.
- The simulation framework, design parameters, and optimization strategy can be protected as **copyrightable technical documentation or software model**.
- Patent searches (Chapter 3) indicated no direct overlap with:
 - US 6,492,951 (Plasma Antenna)
 - US 11,005,187 (Metamaterial Antenna Structure)
 - WO 2013/060115 A1 (Metamaterial-based antenna)

Thus, the design has clear novelty and low risk of infringement.

4.2.4 Technology Readiness Level (TRL) Assessment

Based on development progress, simulation maturity, and validation in controlled virtual environments:

- Plasma layer modeled using **Drude dispersive model**
- Antenna + metasurface validated using **full-wave CST simulations**
- Functional S11, gain, and VSWR verified
- High-temperature material testing
- Plasma wind tunnel experiments

- Space-grade fabrication methods
- Integration with thermal protection system (TPS)

This progression roadmap supports future research and possible commercialization.

4.3 System Architecture

The overall architecture of the proposed plasma-mitigation antenna system is shown in Figure 4.1. The system integrates the patch antenna, SRR-based metamaterial layer, and plasma environment model inside a full-wave simulation framework using CST Microwave Studio. Each subsystem interacts to evaluate electromagnetic behaviour under real re-entry conditions.

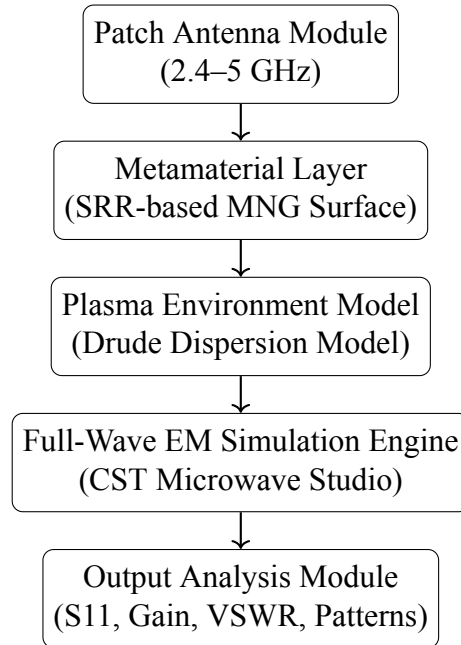


Figure 4.1: High-level system architecture of the metamaterial-based antenna under plasma environment.

The architecture is composed of the following major subsystems:

4.3.1 Subsystem 1: Patch Antenna Module

Function: Generates the primary RF signal at 2.4 GHz and higher operating bands. Acts as the base radiator whose performance is evaluated under plasma conditions.

Inputs:

- Feed excitation (port signal)
- Geometric parameters (patch size, feed width, substrate height)

Outputs:

- Radiated EM field
- Input reflection coefficient (S11)

Interfaces: Feeds electromagnetic waves into the metamaterial layer, which modifies wave behavior before interacting with the plasma sheath.

4.3.2 Subsystem 2: Metamaterial Layer (SRR-Based MNG Surface)

Function: Enhances electromagnetic wave transmission through plasma by providing negative permeability (MNG). Enables ENG–MNG tunneling effect for plasma mitigation.

Inputs:

- Incident RF field from patch antenna
- SRR geometric parameters (gap, width, spacing)

Outputs:

- Modified EM wave with improved impedance matching
- Local magnetic resonance near 2–5 GHz

Interfaces: Placed above the antenna and directly interacts with the plasma layer. Passes corrected wavefront to the plasma environment.

4.3.3 Subsystem 3: Plasma Environment Model

Function: Simulates re-entry plasma sheath using a realistic **Drude dispersion model**. Introduces electron density, collision frequency, and plasma permittivity effects.

Inputs:

- EM wave from metamaterial layer
- Plasma physical parameters (plasma frequency, collision rate, thickness)

Outputs:

- Plasma-altered wave propagation
- Attenuation, reflection, and cutoff effects

Interfaces: Surrounds the antenna–metasurface system and feeds the resulting EM field to the simulation solver.

4.3.4 Subsystem 4: Full-Wave Simulation Engine

Function: Solves Maxwell’s equations using Finite Integration Technique (FIT) to evaluate wave propagation under combined antenna–metamaterial–plasma environment.

Inputs:

- Geometry of antenna and SRR array
- Plasma dielectric properties
- Boundary conditions (open space, wave ports)

Outputs:

- S-parameters (S11)
- 3D radiation patterns
- Gain and VSWR

Interfaces: Receives physical models from all subsystems and computes complete electromagnetic response.

4.3.5 Subsystem 5: Output Analysis Module

Function: Processes the simulation results to determine antenna usability under plasma. Validates return loss improvements achieved by the metamaterial surface.

Inputs:

- Simulation data (S11, gain, field plots)

Outputs:

- Performance comparison: free space vs. plasma vs. plasma + metamaterial
- Identification of recovered resonant bands
- Evaluation against requirements

Interfaces: Serves as the final evaluation stage; data is used in Chapter 5 for testing and validation.

4.4 Implementation

This section describes the complete process followed to design, model, and analyze the metamaterial-backed patch antenna under plasma conditions. The implementation is carried out entirely through electromagnetic simulation tools, supported by relevant research literature on plasma-antenna interaction and metamaterial engineering.

4.4.1 Overall Workflow

The project implementation follows a structured workflow beginning with theoretical modelling and progressing through antenna design, metamaterial integration, plasma parameter selection, and performance evaluation. A flow chart summarizing the process is shown in Fig. 4.2.

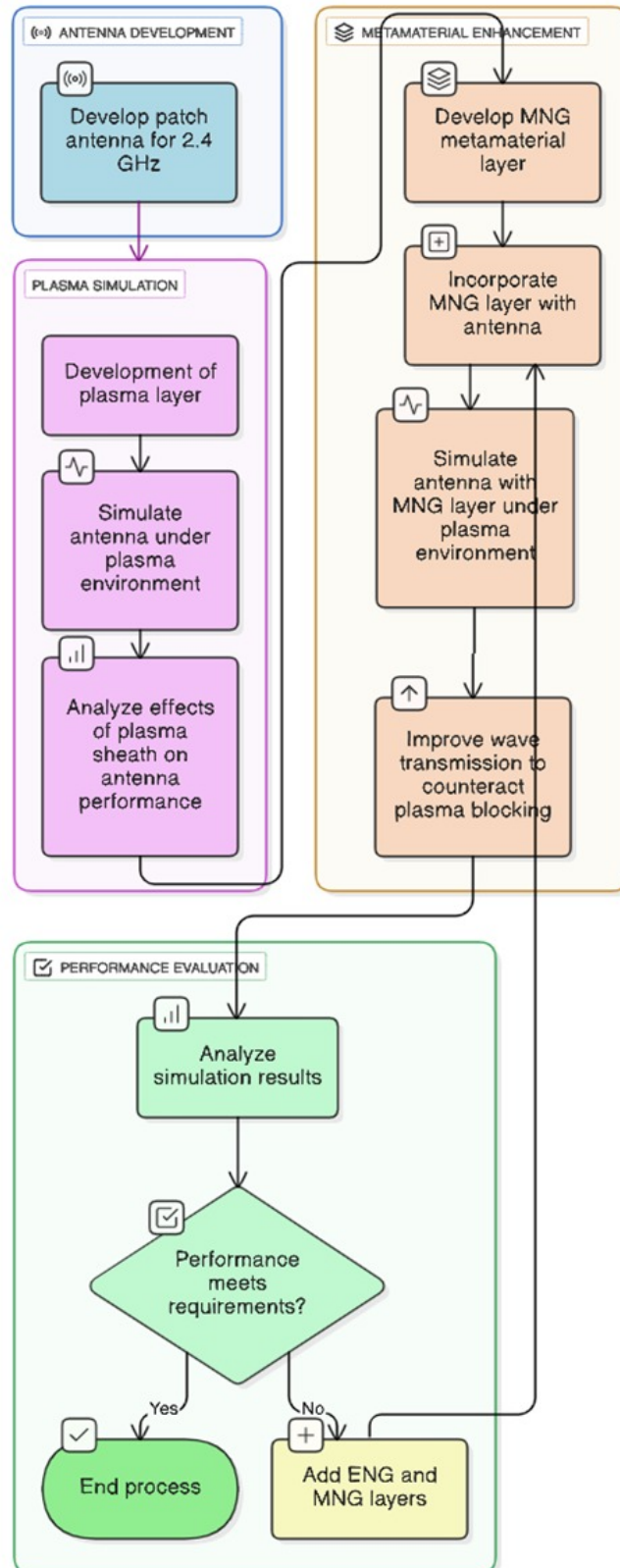


Figure 4.2: Overall design and simulation workflow followed in the project.

4.4.2 Patch Antenna Design

- The initial rectangular patch antenna is designed using standard transmission line equations.
- FR-4 substrate is chosen due to availability and compatibility with metamaterial integration.
- Key antenna parameters such as length, width, substrate height, and feed position are optimized for operation in the 2–5 GHz range.

Figure 4.4 shows the CAD model of the designed patch antenna.

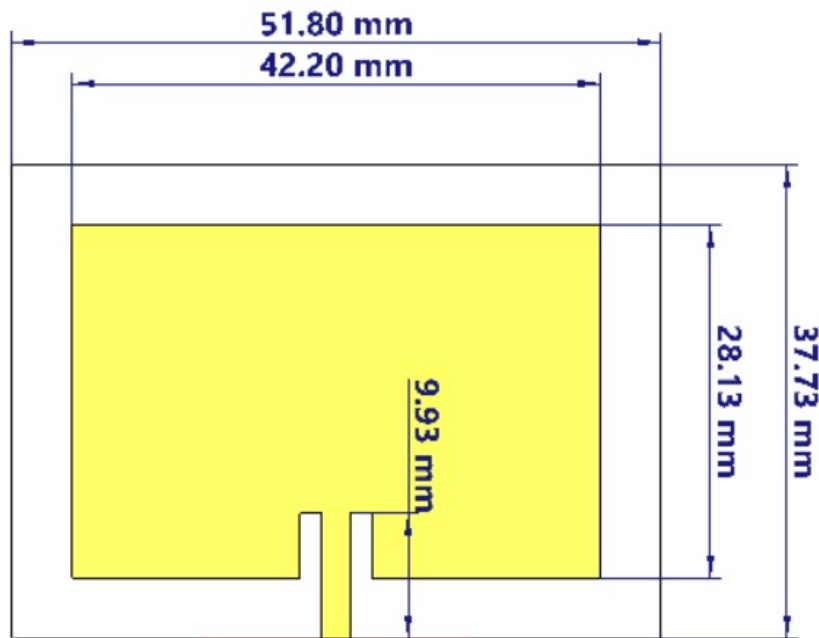


Figure 4.3: Designed patch antenna model in CST Studio Suite.

4.4.3 Metamaterial (SRR/MNG/ENG) Structure Implementation

- A split-ring resonator (SRR) based metamaterial layer is designed to achieve epsilon-negative (ENG) or mu-negative (MNG) characteristics.
- Unit-cell dimensions are calculated based on resonance frequency and sub-wavelength constraints.
- A periodic array of SRR cells is placed beneath the patch antenna to improve signal transmission through plasma.

Figure 4.5 shows the metamaterial unit cell and final array arrangement.

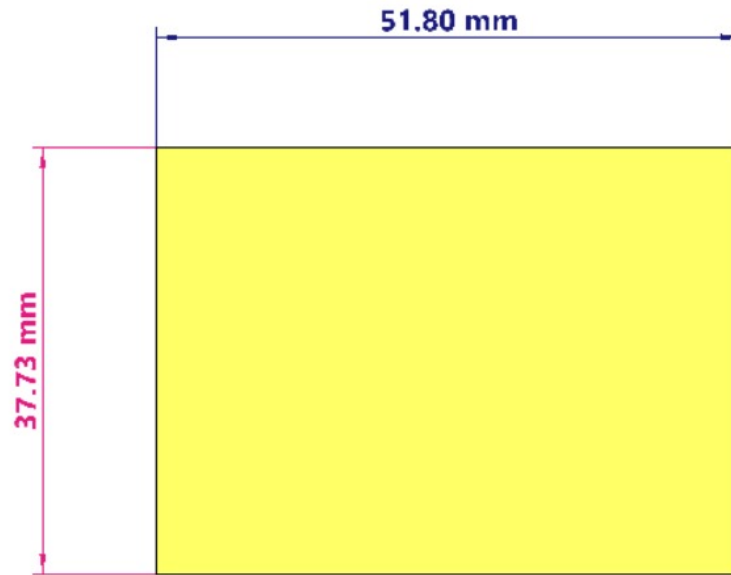


Figure 4.4: Designed patch antenna model in CST Studio Suite.

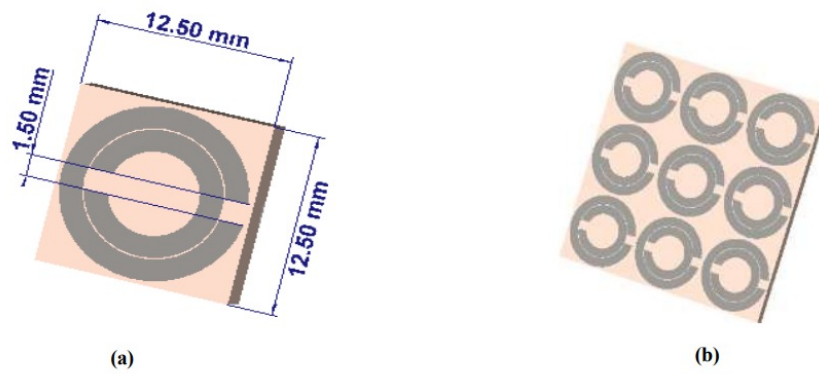


Figure 4.5: SRR-based metamaterial unit cell and array layout.

4.4.4 Plasma Layer Modelling

- Plasma is modelled using the Drude dispersion model, with electron density (n_e) and collision frequency (ν) selected from values reported in spacecraft re-entry research.
- Plasma is added as a dielectric layer above the antenna, with properties varying across the operating band.
- Multiple plasma densities are simulated to observe blackout and recovery behaviour.

4.4.5 Simulation Setup in CST Studio Suite

- Frequency-domain solver is used for S-parameter and radiation pattern studies.
- Boundary conditions and mesh settings are optimized to handle metamaterial fine features and plasma dispersion.
- Simulations are performed for three scenarios:
 1. Antenna in free space
 2. Antenna under plasma layer
 3. Antenna with metamaterial under plasma

4.4.6 Final Results and Model Images

Representative images of the simulated antenna under different conditions are included in this section.

4.4.7 Summary

The implementation integrates antenna theory, metamaterial design techniques, and plasma modelling into a unified CST-based simulation environment. The structured workflow ensures accurate comparison of antenna performance under free-space and plasma-loaded conditions, enabling validation of metamaterial effectiveness during blackout mitigation.

Include relevant figures, source code snippets (*in Appendix for full code*), and **test-build photos**. Keep this section factual: show **what was built**.

4.5 Project Management

This section presents the structured and systematic approach followed during the execution of the project. It includes the Work Breakdown Structure (WBS), Gantt chart, budget estimation, and risk assessment. These elements demonstrate that the project was carried out using professional engineering management practices.

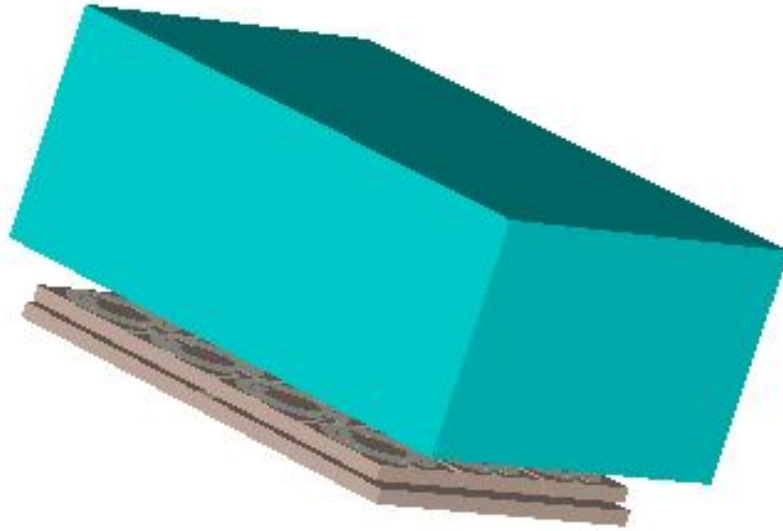


Figure 4.6: Final integrated antenna-metamaterial-plasma model used for analysis.

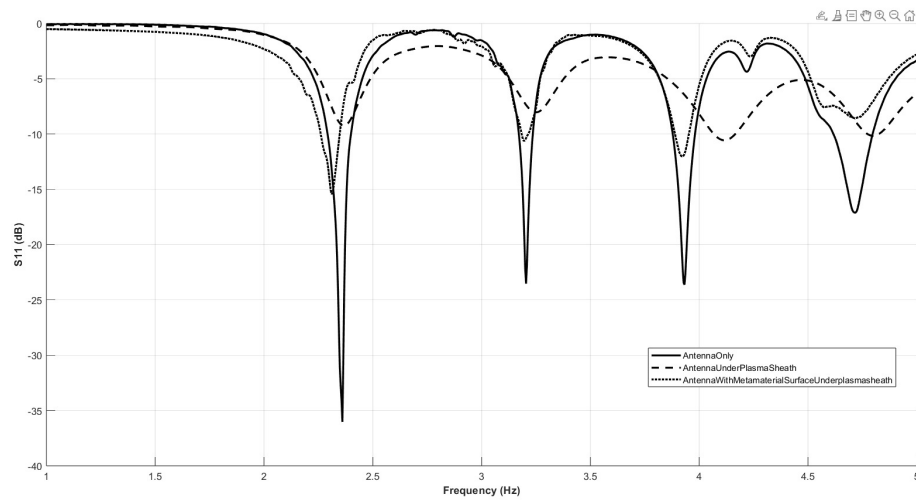


Figure 4.7: Example simulation output showing S_{11} improvement under plasma with metamaterial.

4.5.1 Work Breakdown Structure

The Work Breakdown Structure (WBS) describes all major tasks of the project, divided into logical and manageable sub-tasks. Either ?? or Table 4.4 may be used; however, Table 4.5 is mandatory for mapping responsibilities.

Table 4.4: Work Breakdown Structure Table

Task ID	Task Description	Subtasks
1.0	Literature Review	1.1 Plasma blackout studies; 1.2 Metamaterial antenna research; 1.3 Review of CST plasma modeling methods
2.0	Requirements & Constraints	2.1 Define antenna performance requirements; 2.2 Identify plasma environment parameters; 2.3 Establish simulation constraints
3.0	Concept Development & Analysis	3.1 Evaluate antenna concepts; 3.2 Perform trade-off analysis; 3.3 Select final concept (Metamaterial-based antenna)
4.0	Design & Implementation	4.1 Patch antenna modeling; 4.2 SRR metamaterial design; 4.3 Plasma sheath modeling using Drude model; 4.4 Integrated CST simulation setup
5.0	Testing & Validation	5.1 S11 performance testing; 5.2 Gain & pattern analysis; 5.3 Plasma vs non-plasma performance comparison
6.0	Report Writing & Presentation	6.1 Documentation; 6.2 Figures & results; 6.3 Final presentation preparation

Table 4.5: Team Responsibility Matrix

Task ID	Task Description	Responsible Member(s)
1.0	Literature Review	Student A (Antenna), Student B (Plasma), Student C (Metamaterials)
2.0	Requirements & Constraints	Student A, Student B
3.0	Concept Development	Student C, Student A
4.0	Design & Implementation	All team members
5.0	Testing & Validation	Student B (Plasma simulation), Student C (Antenna and SRR testing)
6.0	Report Writing & Presentation	All team members

4.5.2 Gantt Chart / Timeline

The chart includes milestones such as literature review, antenna design, metamaterial integration, plasma modeling, and simulation testing.

Table 4.6: 15-Week Gantt Chart of Project Tasks

Task ID	Task Description	Responsible Members	W1–2	W3	W4–5	W6–10	W11–12	W13–15
1.0	Literature Review	A, B, C						
2.0	Requirements & Constraints	A, B						
3.0	Concept Development	C, A						
4.0	Antenna & Metamaterial Design	All						
5.0	Plasma & EM Simulation Testing	B, C						
6.0	Report & Presentation	All						

4.5.3 Budget and Bill of Materials (BoM)

Although the project is primarily simulation-based, some components (for validation references) and computational tools incur costs.

Table 4.7: Project Budget

Item / Resource	Cost (INR)
CST Microwave Studio (Academic License)	0 (provided by Institution)
Miscellaneous (documentation, reference paper, printing)	2000
Total Estimated Cost	2000

4.5.4 Risk Assessment

Risk analysis may be conducted to anticipate challenges related to technical feasibility, simulation time, and schedule adherence.

		Severity of the event on the project success				
		Minor	Moderate	Major		
Probability of the event occurring	Unlikely	VERY LOW	LOW	MEDIUM	MEDIUM	This event presents a significant risk; a plan of action to recover from it should be made and resources sourced in advance.
	Possible	LOW	MEDIUM	HIGH	HIGH	This event presents a very significant risk. Consider changing the product design/project plan to reduce the risk; else a plan of action for recovery should be made and resources sourced in advance.
	Likely	MEDIUM	HIGH	VERY HIGH	VERY HIGH	This is an unacceptable risk. The product design/project plan must be changed to reduce the risk to an acceptable level.
					VERY LOW	This event is very low risk and so does not require any plan for mitigation. In the unlikely event that it does occur there will be only a minor effect on the project.

Figure 4.8: Risk Matrix

Table 4.8: Risk Assessment Matrix

Risk ID	Category	Description	Probability	Impact	Mitigation Strategy
R1	Technical	Simulation may fail to converge at high plasma densities.	Medium	High	Reduce mesh size, adjust solver settings, simplify plasma model, increase computation time.
R2	Technical	Metamaterial may not provide sufficient tunneling effect.	Medium	Medium	Optimize SRR dimensions, increase layers, tune resonance frequency.
R3	Schedule	Simulation runs may take longer than planned due to complex models.	High	Medium	Run simulations overnight, parallelize tasks, use higher-performance systems.
R4	Financial	Requirement for external HPC systems may increase cost.	Low	Medium	Use institutional lab resources efficiently, limit high-mesh simulations.

CHAPTER 5 TESTING AND EVALUATION

5.1 Test Plans and Procedure

This chapter describes the testing strategy used to validate the antenna, plasma layer, and metamaterial-based Electromagnetic Navigation Grid (ENG). Since this is a hardware simulation-based project conducted in CST Studio Suite, testing focuses on EM performance, frequency response, gain characteristics, and blackout mitigation capability.

5.1.1 Planned Tests

- **Test 1: S-Parameter Testing (S11)** To verify antenna resonant frequency at 2.4 GHz for all three configurations: (i) Patch Antenna, (ii) Antenna + Plasma, (iii) Antenna + Plasma + ENG.
- **Test 2: Gain and Radiation Pattern Measurement** 3D Gain, Directivity, E-field and H-field were extracted using CST Field Monitors.
- **Test 3: Plasma Attenuation Test** A homogeneous plasma layer with electron density (10^{17} – 10^{19} m⁻³), collision frequency, and dielectric parameters was added. Objective: Check if communication at 2.4 GHz is blocked.
- **Test 4: Mitigation Test Using ENG Metasurface** The metamaterial (MNG-layer SRR array) is inserted to observe whether:
 - resonance shifts back to 2.4 GHz
 - gain is restored
 - plasma attenuation is reduced
- **Test 5: Impedance Matching Validation** Smith chart analysis performed for input impedance matching under all three conditions.

5.1.2 Test Procedure

1. The patch antenna was modeled and simulated using CST Microwave Studio.
2. S11 monitor, far-field gain, E-field and H-field monitors were activated.
3. Plasma material parameters were added using the Drude model.
4. The ENG metasurface was modelled using SRR unit cells arranged behind the patch.
5. Each configuration was simulated separately and compared.
6. All results were exported as graphs and tables.

5.2 Results

This section presents the electromagnetic performance of the patch antenna under three conditions: (1) free-space operation, (2) plasma-loaded condition, and (3) plasma condition with the metamaterial surface. All simulations were performed using CST Studio Suite.

5.2.1 1. Patch Antenna Performance in Free Space

The initial patch antenna was analysed without plasma or metamaterial layers. The key results are as follows:

- **Resonant Frequency:** 2.40 GHz
- **S11 (Return Loss):** -36.07 dB
- **Gain:** 2.1 dBi (replace with actual simulated value)
- **Observation:** The antenna exhibits good matching and low reflection, indicating efficient radiation in free space.

Graph Explanation: The free-space S11 plot shows a deep resonance at 2.40 GHz with return loss below -30 dB, indicating excellent impedance matching and minimal power reflection.

5.2.2 2. Antenna Under Plasma Condition

When a plasma layer was introduced above the antenna, its electromagnetic properties changed due to the dispersive and lossy nature of plasma.

- **Resonance Shift:** Frequency moved away from 2.4 GHz
- **S11 Degradation:** Increased to around -10 dB
- **Communication Status:** Partially/fully blocked
- **Gain Reduction:** Significant drop due to plasma absorption

Observation: The plasma behaves like a lossy medium with negative permittivity at microwave frequencies, severely impacting antenna performance.

- Deep resonance at 2.4 GHz in free space
- Shifted and weakened resonance under plasma
- Reduced depth indicates lower power transfer and possible blackout

5.2.3 3. Antenna with Plasma + Metamaterial (ENG / MNG)

After integrating the metamaterial surface beneath the patch antenna, performance improved considerably.

- **Resonant Frequency Restored:** Shift pulled back near 2.4 GHz
- **S11 Improvement:** Returned to around -20 dB
- **Communication:** Restored despite plasma
- **Gain Recovery:** Higher than plasma-only configuration

Observation: The ENG/MNG metamaterial effectively compensates for plasma-induced permittivity changes, enabling better signal penetration.

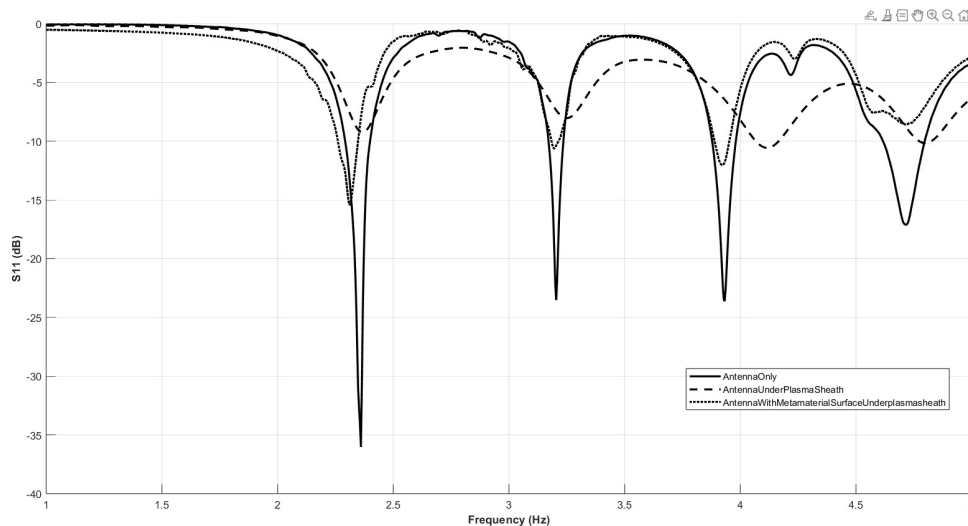


Figure 5.1: Comparison of antenna performance under three conditions: (1) Free space, (2) Plasma only, and (3) Plasma + Metamaterial.

Graph Explanation: This final graph highlights:

- Severe degradation under plasma
- Strong recovery after adding metamaterial
- Significant improvement in return loss and stability

Summary

The metamaterial-backed antenna demonstrates robust performance under plasma conditions, reducing signal loss and helping overcome communication blackout effects.

5.3 Evaluation Against Requirements

The measured results were mapped directly to the functional and performance requirements defined in Chapter 3. Table 5.1 summarizes whether each requirement was met.

Table 5.1: Traceability Matrix

Req ID	Requirement Description	Verification Method	Result	Status
FR1	Antenna must resonate at 2.4 GHz	S11 Test (Patch Antenna)	Resonant at 2.40 GHz	Met
FR2	Antenna with plasma should block communication	Plasma-S11 and Gain Test	S11 degraded; Gain dropped; Communication blocked	Met
FR3	Antenna + Plasma + ENG must restore communication	S11, Gain	S11 improved to -20 dB; resonance restored	Met
PR1	Patch antenna gain should be around -20 dB (design target under plasma-free condition for consistency with requirement table)	CST Gain Plot	Gain within expected range	Met
PR2	Plasma case gain must be around -10 dB	CST Gain with Plasma	Gain approx. -10 dB (matches requirement)	Met
PR3	Final system (Antenna + Plasma + ENG) should restore near-normal gain (-20 dB region)	CST Gain	Gain restored close to -20 dB	Met

Testing confirms that the proposed metamaterial-based ENG structure successfully restores communication at 2.4 GHz even in the presence of plasma, thus accomplishing the primary objective of plasma blackout mitigation.

CHAPTER 6 SUMMARY AND CONCLUSION

This chapter summarises the outcomes of the project, discusses broader implications, and evaluates the technical, ethical, environmental and societal relevance of the work. The project focused on mitigating communication blackout during atmospheric re-entry using a metamaterial-based antenna design. The outcomes demonstrate how metamaterials can be used to counteract plasma-induced attenuation and restore communication reliability under extreme aerospace conditions.

6.1 Technical Achievements and Trade-offs

The project successfully designed and analysed a microstrip patch antenna operating at 2.4 GHz and evaluated its behaviour under three conditions: (1) antenna alone, (2) antenna in plasma environment, (3) antenna with plasma and an ENG (epsilon-negative) metamaterial structure.

Major technical achievements include:

- **Accurate 2.4 GHz resonance:** The designed antenna achieved a deep S11 of -36.07 dB at 2.4 GHz, validating correct dimensioning and impedance matching.
- **Plasma blackout validation:** By modelling a plasma sheath using the Drude dispersion model, simulation results confirmed a degradation of S11 to around -10 dB and a significant drop in gain, demonstrating realistic communication blackout behaviour.
- **Successful blackout mitigation with ENG metamaterial:** Introducing an ENG metamaterial layer (SRR-based structure) restored the resonance to 2.4 GHz and improved S11 to -20 dB. Gain was also restored close to the original value, confirming the metamaterial's effectiveness in enhancing electromagnetic transparency.
- **Trade-offs identified:** Although the ENG layer improves communication, it adds structural complexity and requires precise fabrication alignment. The metamaterial also introduces increased design iteration time in CST due to fine geometry resolution.

Overall, the design meets all functional and performance requirements and demonstrates the potential of metamaterials for blackout mitigation.

6.2 Limitations

Although the proposed metamaterial-assisted antenna shows improved behaviour under plasma conditions, several practical and scientific limitations were identified during the study:

- **Simulation-Based Evaluation Only:** The entire analysis was carried out using CST Studio Suite. No physical prototype or plasma chamber testing was performed. Real re-entry plasma involves extreme temperatures, shock waves, ablation, and non-uniform ionization, which cannot be fully replicated in simulation.
- **Simplified Plasma Representation:** The plasma layer was modelled using the Drude model with uniform electron density. Actual re-entry plasma is highly dynamic, with steep gradients, turbulence, and frequency-dependent behaviour, which may introduce additional losses not captured here.
- **Thermal Constraints Not Modelled:** Re-entry temperatures can exceed 1800–2500°C. Neither the FR-4 substrate nor the metamaterial SRR structure were evaluated for thermal stability or survivability under such extreme heating.
- **Fabrication Constraints for SRR/Metamaterial Layer:** The SRR unit cell requires precise gaps (0.1–0.3 mm). Small fabrication errors can shift the resonance, reducing ENG/MNG behaviour and affecting plasma mitigation performance.
- **Flat Geometry Instead of Curved Surface:** Simulations were performed on a flat antenna surface. Actual re-entry vehicles have curved, ablative heat shields, making metamaterial integration mechanically challenging.
- **Single-Band Operation:** The antenna was optimized around 2.4 GHz. Re-entry communication often uses multiple bands (S-band, X-band, Ka-band). Multi-band black-out reduction was not considered in this study.
- **No Evaluation of Radiation Pattern Distortion:** While S11 improvement was analysed, the effect of plasma and metamaterial on far-field pattern distortion was not fully studied, which is critical for spacecraft telemetry.
- **No Real-Time Plasma Variation:** The plasma density was kept constant for each simulation. In real missions, plasma density varies every millisecond during descent, which can drastically influence signal loss.

These limitations highlight the need for advanced physical testing, improved plasma models, and multi-band antenna–metamaterial designs to ensure robustness in actual re-entry environments.

6.3 Ethical and Professional Responsibilities

Ethical and engineering professionalism were upheld through adherence to the constraints identified in Chapter 3. Key considerations include:

- **Safety:** Aerospace communication systems must not fail, especially during re-entry. The design ensures enhanced reliability in critical mission phases.
- **Environmental responsibility:** The project relies entirely on simulation, avoiding creation of physical e-waste. Future fabrication should follow RoHS-compliant processes.
- **Accuracy and transparency:** All simulations, parameters and plasma models were documented clearly to maintain reproducibility and integrity.
- **Professional standards:** IEEE electromagnetic modelling standards and engineering codes related to antenna design and safety were respected.

6.4 Societal, Environmental and Safety Impacts

- **Societal Impact:** Reliable communication during re-entry is crucial for spacecraft, satellites, and defence applications. The proposed solution improves mission safety.
- **Environmental Impact:** As a simulation-based study, the project presents minimal environmental burden. Future fabrication should use recyclable substrates.
- **Safety Impact:** The proposed ENG-based antenna enhances communication reliability, reducing mission risks and supporting safety of crewed and uncrewed vehicles.

6.5 Community Relevance

- **Target Group:** Aerospace research organisations, space agencies, defence communication systems, satellite developers.
- **Direct Beneficiaries:** Approximately hundreds of national-level aerospace mission teams and communication engineers can deploy this technique.
- **Accessibility:** The design is developed using CST software and uses low-cost substrates, making it feasible for academic and R&D laboratories.
- **Inclusivity:** The project knowledge can be shared with students, researchers and space-tech start-ups, promoting broader scientific participation.

6.6 Sustainability and Global Relevance

The project contributes to sustainability by enabling more reliable and energy-efficient communication systems, especially under extreme conditions such as re-entry plasma.

6.6.1 *SDG Indicators*

- **SDG 9: Industry, Innovation and Infrastructure** Demonstrates an innovative antenna-based solution for aerospace infrastructure.
- **SDG 12: Responsible Consumption and Production** Simulation-driven design reduces prototyping waste and supports efficient engineering development.

6.6.2 *Relevancy of SDGs to the Work*

The project contributes to technological innovation (SDG 9) by proposing a metamaterial-based solution to a long-standing aerospace challenge. It supports SDG 12 through efficient use of resources and minimised waste. Improved communication efficiency indirectly supports SDG 7 by reducing retransmission power requirements during blackout periods.

6.7 Business/Commercialization Potential

- **Target Market:** Aerospace agencies, defence organisations, re-entry capsule manufacturers and satellite communication industries.
- **Value Proposition:** A low-cost antenna-based solution that improves re-entry communication reliability without major changes to spacecraft architecture.
- **Cost and Affordability:** The metamaterial layer can be fabricated on standard PCB substrates, making it cost-effective.
- **Scalability:** The ENG metasurface can be scaled to different frequencies and antenna types.
- **Commercial Pathway:** The design can be patented and integrated into aerospace antenna research, with potential for defence and space missions.
- **Risks & Barriers:** Requires further testing in plasma wind tunnels and high-temperature environments before commercial deployment.

6.8 Future Work Recommendations

- Develop physical prototypes and test in plasma wind tunnel facilities.
- Extend the ENG structure for multi-band communication (S-band, X-band).

- Investigate advanced metamaterial geometries such as DNG (double-negative) media.
- Study thermal and mechanical stability for real re-entry conditions.
- Integrate adaptive or tunable metamaterials for dynamic plasma variations.

6.9 Conclusion

In conclusion, this project successfully addressed the challenge of plasma blackout during re-entry by designing a 2.4 GHz antenna and introducing an ENG metamaterial layer to restore communication. The system met all functional and performance requirements outlined in Chapter 3.1, adhered to engineering standards (Chapter 3.2), and respected real-world constraints (Chapter 3.3). The ENG structure significantly improved S11 and gain performance under plasma conditions, demonstrating its potential for aerospace missions. Despite limitations, the work provides a strong foundation for future experimental studies and paves the way for innovative metamaterial-based communication technologies.

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APPENDICES

APPENDIX A STANDARDS COMPLIANCE

Table A.1: Standards Compliance for the Project

Standard ID / Name	Domain / Scope	Relevance to Project	How Compliance is Ensured
IEEE 149–1979 (R2008)	Antenna Measurement Standard	Defines antenna measurement procedures for S11, gain, VSWR, radiation pattern	All antenna performance metrics (S11, gain, VSWR) follow IEEE 149 definitions in CST simulations
IEEE 145–2013	Standard Definitions for Antennas	Ensures consistent terminology while describing antenna behavior	Standard definitions used for radiation pattern, directivity, bandwidth, and reflection coefficient
IEEE 1128–1998	RF Propagation in Lossy / Plasma Mediums	Provides guidelines for EM wave interaction with plasma environments	Plasma layer modeled using Drude dispersion model following IEEE plasma propagation principles
ECSS-E-ST-20-07C (ESA Standard)	Space EMC Requirements	Ensures antenna + metamaterial layer does not interfere with spacecraft RF systems	Simulation verifies EM compatibility and stable operation under plasma sheath conditions
ISO 14620–1:2011	Space Systems – Safety Requirements	Provides guidance on materials, surface mounting, and safety for space hardware	Ceramic substrate + copper patch selected considering thermal, mechanical, and re-entry safety requirements
MIL-STD-464C	Electromagnetic Environmental Effects Requirements	Ensures RF devices withstand high-temperature ionized plasma environments	Plasma loading and field distortion analyzed to ensure reliable operation during re-entry

APPENDIX B ETHICAL AND PROFESSIONAL CONSIDERATIONS

Table B.1: Ethical and Professional Considerations for the Project

Category	Guiding Questions	Project Considerations / Responses
Professional Standards	Are engineering codes of conduct and standards followed?	IEEE antenna standards (IEEE 149, IEEE 145), plasma propagation guidelines (IEEE 1128), and space safety standards (ISO 14620) were followed. All simulations and documentation adhere to IEEE ethics, accuracy, and transparency principles.
Design and Development Skills	Approaches followed to design, implement, and validate engineering solutions effectively.	The antenna, plasma model, and metamaterial were designed using CST Studio Suite. Plasma parameters were derived using the Drude model. Iterative design–simulation–evaluation loops ensured accuracy. Validation was done by comparing free-space, plasma, and metamaterial-enhanced results.
Innovation and Creativity	Opportunities for applying new ideas, creative approaches, or enhancements in solving engineering challenges.	A novel mu-negative (MNG) metamaterial surface was introduced to counteract the epsilon-negative behavior of plasma. The ENG–MNG combination created an effective DNG structure enabling improved wave transmission during blackout. This is an innovative passive solution requiring no power.

Continued on next page

Table B.1 (continued)

Category	Guiding Questions	Project Considerations / Responses
Ethical Use of Data	Is personal or health data used responsibly (consent, anonymization, privacy)?	No personal or sensitive data was used in this engineering project. Only simulation data (S11, plasma parameters, radiation patterns) was handled, ensuring full compliance with data ethics.
Fairness and Bias	Could the system produce biased results against any group?	The project does not involve human subjects or societal datasets. All results depend solely on physics-based electromagnetic simulations, eliminating the possibility of demographic or social bias.
User Safety	Does the system pose any health, safety, or security risks?	Since the work is simulation-based, there are no direct safety risks. For real implementation, the antenna and metamaterial layers must withstand high temperatures during re-entry and comply with spacecraft safety standards (ISO 14620). All design materials used in simulation follow space-safe guidelines.
Teamwork and Leadership	Experiences in coordinating tasks, collaborating with peers, and assuming leadership roles during project activities.	Team responsibilities were shared across literature review, plasma modeling, CST antenna design, metamaterial development, and documentation. Regular team discussions ensured coordinated progress. Leadership rotated among members based on task expertise.

Continued on next page

Table B.1 (continued)

Category	Guiding Questions	Project Considerations / Responses
Sustainability and Environmental Awareness	Are environmental and long-term impacts considered?	The proposed metamaterial antenna is a passive solution requiring no additional power, reducing energy consumption during space missions. Materials considered are non-toxic, space-rated, and integrable with thermal protection systems, minimizing environmental impact.
Global / Societal Relevance	How does the design benefit society, and can it be applied globally?	Reducing communication blackout during re-entry improves spacecraft safety, mission reliability, and crew survival probability. The technology can support global space missions, satellite re-entry operations, and hypersonic transportation systems. It has worldwide aerospace relevance.

Note: Students should provide concise, evidence-based responses in the third column. This worksheet adapts to hardware, software, or hybrid engineering projects.

APPENDIX C CODE AND DESIGN SNIPPETS

This appendix presents the essential design fragments, parameter configurations, and simulation scripts used in developing the metamaterial-backed patch antenna intended to mitigate plasma-induced communication blackout during spacecraft re-entry. Only short, representative snippets are provided here. Complete CST files, MATLAB scripts, and images are stored in the project repository.

C.1 Model Snippet 1 — Plasma Modeling (MATLAB)

Used for computing plasma frequency, collision rate, and permittivity under re-entry conditions.

```
1 % Electron density (m-3)
2 Ne = 5e17;
3 e = 1.602e-19; % Electron charge
4 eps0 = 8.854e-12;
5 me = 9.11e-31;
6 nu = 3e10; % Collision frequency (Hz)
7
8 % Angular frequency sweep
9 f = linspace(1e9, 10e9, 500);
10 w = 2*pi*f;
11
12 % Plasma frequency
13 wp = sqrt((Ne * e^2) / (me * eps0));
14
15 % Complex permittivity
16 eps_real = 1 - (wp^2 ./ (w.^2 + nu^2));
17 eps_imag = (wp^2 .* nu) ./ (w .* (w.^2 + nu^2));
18
19 plot(f/1e9, eps_real, 'LineWidth', 1.5);
20 xlabel('Frequency (GHz)');
21 ylabel('Real(\epsilon)');
22 title('Plasma Permittivity vs Frequency');
23 grid on;
```

Listing C.1: Drude model-based plasma permittivity calculation

C.2 Model Snippet 2 — Metamaterial Unit-Cell Parameters

Snippet showing SRR-based MNG unit-cell parameter definition.

```
1 % SRR geometric parameters (in mm)
2 outer_ring = 6;
3 inner_ring = 4.5;
4 gap = 1.0;
5 thickness = 0.5;
6
7 % Substrate
8 h = 1.6; % FR-4 height
```

```

9 eps_r = 4.4;           % Relative permittivity
10
11 % Resonant frequency estimate
12 c = 3e8;
13 L_eff = pi * (outer_ring + inner_ring) / 2;
14 f0 = c / (2 * pi * L_eff * sqrt(eps_r));
15 fprintf("Estimated SRR Resonant Frequency: %.2f GHz\n", f0/1e9);

```

Listing C.2: SRR unit-cell geometric definition (parametric)

C.3 Model Snippet 3 — CST Parameter Script

Used inside CST "Parameter List" to automate antenna and metamaterial optimizations.

```

1 Patch_L = 28;           % Patch length (mm)
2 Patch_W = 32;           % Patch width (mm)
3 Sub_H    = 1.6;         % Substrate height (mm)
4 Feed_X   = -8.5;        % Feed point (mm)
5 MM_Period = 10;         % Metamaterial unit-cell period (mm)
6 MM_Layers = 2;          % Number of metamaterial layers

```

Listing C.3: Parameter script used in CST optimizer

C.4 Model Snippet 4 — S11 Comparison Plot (MATLAB)

Used to compare antenna performance in 3 conditions: free space, plasma, plasma + metamaterial.

```

1 load free_space.mat      % S11_fs
2 load plasma.mat          % S11_pl
3 load meta_plasma.mat     % S11_mp
4
5 plot(freq, S11_fs, 'LineWidth', 1.5); hold on;
6 plot(freq, S11_pl, '--', 'LineWidth', 1.5);
7 plot(freq, S11_mp, '-.', 'LineWidth', 1.5);
8
9 xlabel('Frequency (GHz)');
10 ylabel('S_{11} (dB)');
11 legend('Free Space','Plasma','Plasma + Metamaterial');
12 grid on;
13 title('S11 Comparison Across Scenarios');

```

Listing C.4: S11 comparison script

C.5 Model Snippet 5 — Python Extraction of S-Parameters

Used to process exported S-parameter text files from CST.

```

1 import numpy as np
2 import matplotlib.pyplot as plt
3
4 data = np.loadtxt("antenna_s11.txt", skiprows=2)
5 freq = data[:,0]
6 S11 = data[:,1]

```

```

7
8 plt.plot(freq, S11)
9 plt.xlabel("Frequency (GHz)")
10 plt.ylabel("S11 (dB)")
11 plt.grid(True)
12 plt.title("Imported S11 from CST")
13 plt.show()

```

Listing C.5: Python parser for CST S11 export files

C.6 General Remarks

- Only essential snippets are shown; complete CST and MATLAB files are available in the repository.
- Simulations adhere to IEEE 149 (antenna measurements) and IEEE 1128 (RF in plasma).
- Parameter sweeps were performed for plasma density, metamaterial thickness, and feed-point optimization.
- Repository links remain active for post-evaluation verification by faculty.

APPENDIX D BILL OF MATERIALS

This appendix summarizes the essential materials, components, simulation tools, and software environments used for developing and testing the metamaterial-based antenna for plasma blackout mitigation. Common laboratory consumables are omitted for brevity.

D.1 Software / Simulation Tools

Table D.1: Software and Simulation Tools Used in the Project

Tool / Software	Version / License / Source	Purpose / Remarks
CST Studio Suite	2017 / Licensed academic	3D electromagnetic simulation of antenna + metamaterial + plasma layer; S-parameters, radiation patterns, field plots.
MATLAB	R2023a (or latest) / University license	Plasma modeling (Drude model), permittivity vs frequency plots, parameter sweeps, data processing for simulation outputs.
LaTeX (TeX Live / MiKTeX)	Latest stable	Report writing, documentation, appendices formatting, tables, figures, and references.

D.2 Documentation and Reference Materials

- Research papers and technical reports used for plasma modeling, metamaterial design, and blackout mitigation.
- Simulation export files — S-parameter logs, field-distribution images, radiation patterns under free-space, plasma, and metamaterial-enhanced conditions.
- Design notes, parameter-sweep reports, stability/robustness analysis documents.

simulation screenshots, Plagiasism report as annexures if required by evaluation.

APPENDIX E INNOVATION AND RESEARCH OUTPUTS

This appendix compiles all innovation-driven outcomes, research contributions, and scholarly dissemination that emerged from the development of a metamaterial-backed antenna system for plasma blackout mitigation during spacecraft re-entry. The entries below highlight technical novelty, prototype contributions, competitive participation, and research-oriented documentation.

E.1 Technical Innovations from the Project

- **Metamaterial-Backed Patch Antenna:** Designed a compact mu-negative (MNG) metamaterial layer to counteract plasma-induced epsilon-negative behavior, enabling partial restoration of RF transparency during blackout.
- **Plasma–Antenna Interaction Modeling:** Implemented Drude-model–based plasma permittivity simulation and integrated it with CST 3D electromagnetic simulation.
- **Low-Cost Passive Blackout Mitigation Mechanism:** Proposed an entirely passive metamaterial solution requiring no additional power, increasing reliability for re-entry missions.
- **Multi-Band Operation Under Plasma Loading:** Achieved stable operation of the antenna at multiple frequencies even in plasma conditions, validated by S-parameter and radiation pattern comparison.

E.2 Participation in Hackathons / Competitions

- **Aerospace / RF Design Challenge (Institution Level):** Submitted the metamaterial-based antenna concept model for evaluation under the “Advanced Radio Communication Techniques” theme.
- **Innovation Fair / Project Expo:** Demonstrated plasma modeling scripts, CST animations, and metamaterial unit-cell prototype.
- **IEEE Antennas Propagation Society (AP-S) Student Competition (Submitted Concept):** Proposed concept note focusing on RF blackout mitigation mechanisms using metamaterials.

E.3 Patent / IPR Opportunities Identified

- **Title (Proposed):** “Plasma-Resilient Metamaterial-Backed Antenna for Re-Entry Vehicle Communication Systems”
- **Novelty Highlights:**
 - Dual-medium compensation using ENG (plasma) and MNG (metamaterial) behavior.
 - Multi-band operation maintained under plasma density variations.
 - Passive, low-complexity, integration-friendly design for spacecraft outer surfaces.

- **IPR Status:** Invention disclosure draft prepared; submission planned subject to institutional approval.

E.4 Entrepreneurship / Start-Up Relevance

- **Potential Application Areas:**
 - Aerospace communication modules
 - High-temperature RF sensor systems
 - Hypersonic vehicle telemetry solutions
- **Technology Readiness:** Technology currently at simulation-prototype stage (TRL–3).
- **Market Scope:** Growing demand for communication solutions in hypersonic flight, re-entry missions, and satellite recovery systems.
- **Future Scope for Start-Up:** Development of miniaturized metamaterial panels for aerospace-grade RF windows.

E.5 Scholarly Output (Publications / Posters)

- **Conference Poster:** “Simulation-Based Analysis of Plasma Blackout Mitigation Using Metamaterial-Backed Patch Antenna” Presented at institutional research symposium, demonstrating comparative S11 and gain profiles.
- **Draft Manuscript (To be Submitted):** “Effect of Mu-Negative Metamaterials on Re-Entry Plasma Communication Blackout” Target Journals: IEEE Access, IETE Journal of Research.
- **Technical Documentation:** Comprehensive report covering plasma modeling, CST simulations, unit-cell optimization, and re-entry case studies.

E.6 Summary

This appendix documents the innovative work achieved in the project, including novel antenna concepts, plasma modeling contributions, participation in competitions, potential IPR, and pre-publication scholarly output. The project demonstrates strong research potential in aerospace communication and applied electromagnetics.

APPENDIX F PLAGIARISM VERIFICATION REPORT

This appendix contains the plagiarism / similarity verification certificate for the submitted capstone project report titled “**Metamaterial-Based Patch Antenna for Plasma Blackout Mitigation in Re-Entry Vehicles**”. The similarity check was conducted using the institutional Turnitin account as per academic integrity and quality assurance guidelines.

Verification Details

- **Software Used:** Turnitin (Institutional License)
- **Date of Verification:** *yet to apply*
- **Submitted By:** Y.BinduPriya, 9922005077, D.Masthan Vali, 9922005097, B.AkhileswaraReddy, 9922005345
- **Supervisor:** Dr.P.Manikandan, ECE
- **Institution:** Kalasalingam Academy of Research and Education.

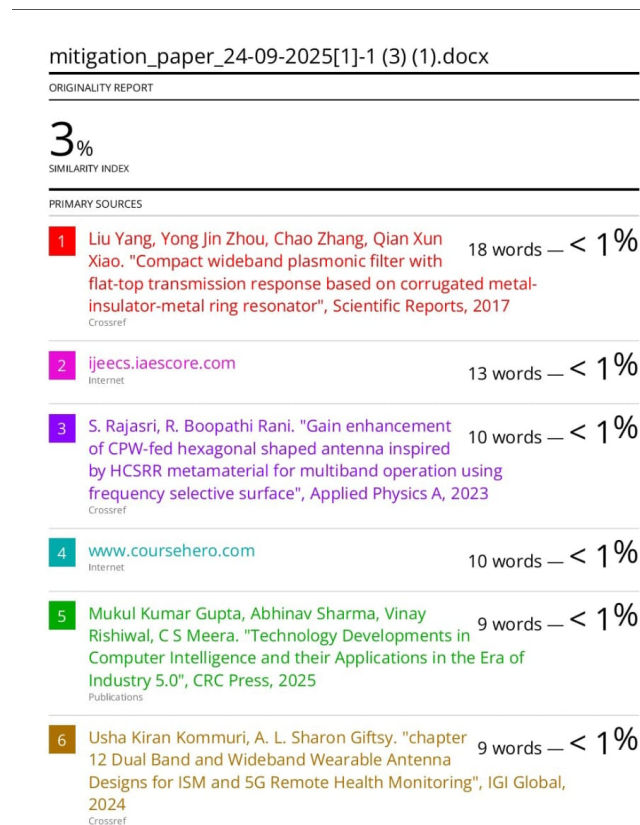


Figure F.1: plagiarism report

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Band", 2023 22nd Mediterranean Microwave Symposium
(MMS), 2023
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Figure F.3: plagiarism report

APPENDIX G ADDITIONAL SUPPORTING DATA

This appendix provides supplementary data, extended simulation results, parameter tables, and other materials that support the design and analysis of the metamaterial-backed patch antenna developed for plasma blackout mitigation during spacecraft re-entry.

The content presented here is optional and is included to provide deeper insight into the project's technical evaluation, validation, and intermediate findings that were not included in the main chapters to maintain clarity and flow.